

Software Requirements Specification

for

*Adaptive Explainable Multi-Agent Retrieval-Augmented Framework for
Intelligent Document Analysis*

Version 1.0 approved

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Project Id : G566

Date Created: September 13, 2026

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4.4.1 Description and Priority

High priority. Enables administrators to manage users, documents, permissions, audit records, and system activity.

4.4.2 Stimulus/Response Sequences

Admin logs in → System verifies role → Admin views authorized records → Admin performs permitted management actions.

4.4.3 Functional Requirements

REQ-15: System shall allow admins to view document and user records.

REQ-16: System shall allow admins to manage user accounts and document access.

REQ-17: System shall restrict admin-only endpoints through RBAC.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

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5.3 Security Requirements

5.4 Software Quality Attributes

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6. Other Requirements

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1. Introduction

1.1 Purpose

This document outlines the software requirements for the Adaptive Explainable Multi-Agent Retrieval-Augmented Framework for Intelligent Document Analysis, a web-based platform for evidence-backed document analysis using adaptive retrieval, multi-agent reasoning, verification, and explainability.

1.2 Document Conventions

All requirement IDs are prefixed with REQ-, and the sections follow an IEEE-style Software Requirements Specification structure.

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, testers, project guides, project managers, researchers, and users who design, implement, validate, or evaluate the system.

1.4 Product Scope

The system provides document upload, natural-language analysis, adaptive BM25/Dense/Hybrid retrieval, Cross-Encoder reranking, multi-agent processing, evidence verification, citations, and confidence-aware results.

1.5 References

Project presentation and abstract; System Architecture Diagram; Class Diagram; Sequence Diagram; State Chart Diagram; Deployment Diagram.

2. Overall Description

2.1 Product Perspective

The framework is a full-stack web application using Next.js/React and FastAPI. Documents are parsed, chunked, embedded, indexed for lexical and vector retrieval, and processed through adaptive retrieval and LangGraph orchestration. PostgreSQL stores structured data and Qdrant stores document embeddings.

2.2 Product Functions

- User registration, authentication, JWT session management, and RBAC
- PDF, DOCX, and TXT upload, parsing, chunking, and metadata extraction
- BGE embeddings and Qdrant vector storage
- Natural-language Q&A, summarization, and document comparison
- Adaptive BM25, Dense, and Hybrid retrieval selection
- Cross-Encoder reranking of retrieved passages
- LangGraph multi-agent orchestration for query, retrieval, reasoning, summarization, comparison, and verification
- Evidence verification, citations, supporting passages, confidence, and explainable responses

2.3 User Classes and Characteristics

- General users: Upload documents, ask questions, summarize content, compare documents, and inspect evidence-backed responses.
- Researchers and analysts: Perform in-depth document analysis, evaluate retrieved evidence, and use citations and confidence information.
- Organization administrators: Manage users and documents, review audit information, and

2.4 Operating Environment

- Web-based application accessible through modern browsers such as Chrome, Firefox, and Edge.
- Next.js/React frontend and FastAPI (Python) backend.
- PostgreSQL for structured records and Qdrant for vector embeddings.
- Dockerized services with cloud or local deployment.

2.5 Design and Implementation Constraints

- Next.js/React and FastAPI shall be used for the application layers.
- System shall support BM25, dense vector, and hybrid retrieval.
- BGE embeddings and Qdrant shall support semantic retrieval.
- LangGraph shall orchestrate multi-agent workflows.
- JWT, RBAC, and provenance metadata shall protect and trace analysis.

2.6 User Documentation

- Online guide for registration, upload, querying, analysis modes, citations, and confidence.
- In-application onboarding and help information.
- Troubleshooting guidance for processing and retrieval errors.

2.7 Assumptions and Dependencies

- Configured LLM/model services remain available within usage limits.
- PostgreSQL and Qdrant services are available.
- Users have internet access and a modern browser.
- Uploaded files use supported formats.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a responsive web interface including login/register, dashboard, document management, upload and processing status, chat/query, Q&A/summarization/comparison modes, and result views with citations, evidence, verification status, and confidence.

The application operates on standard PCs, tablets, and smartphones. No special hardware interfaces are required.

3.3 Software Interfaces

- PostgreSQL: user records, document metadata, chat history, audit records, and analysis data.
- Qdrant: document embeddings and vector similarity search.
- File storage: uploaded source documents and associated files.
- LLM service: answer generation, summarization, comparison, and reasoning.
- BGE/Sentence Transformers: embeddings and Cross-Encoder reranking.
- PDF/DOCX/TXT parsing libraries for text and metadata extraction.
- JWT/Bcrypt: authentication and password protection.
- FastAPI: REST APIs and service orchestration.

3.4 Communications Interfaces

Frontend and backend communication shall use secure HTTPS. JWT access tokens shall be transmitted using secure authorization mechanisms. Backend services communicate with PostgreSQL, Qdrant, file storage, and configured AI/model services through authenticated interfaces.

4. System Features

4.1 User Registration and Authentication

4.1.1 Description and Priority

High priority. Provides secure access control and user management.

4.1.2 Stimulus/Response Sequences

User submits registration/login form → System validates credentials → JWT session is issued → User receives access to authorized functions.

REQ-1: System shall allow new users to register.

REQ-2: System shall allow existing users to log in using email and password.

REQ-3: Passwords shall be protected using secure hashing such as bcrypt.

REQ-4: JWT tokens shall be used for authenticated session management.

REQ-10: Secure session renewal/re-authentication shall be supported.

REQ-11: Logout shall invalidate the active session.

REQ-12: Password reset shall be available.

4.2 Idea Submission and Scoring

4.2.1 Description and Priority

High priority. This core operation accepts documents and queries, retrieves relevant evidence, and returns an analyzed result.

4.2.2 Stimulus/Response Sequences

User uploads a document and submits a query → retrieval strategy is selected → evidence is retrieved and reranked → multi-agent analysis is performed → verified response is returned.

4.2.3 Functional Requirements

REQ-5: System shall allow supported documents and natural-language queries.

REQ-6: System shall analyze the query and select BM25, Dense, or Hybrid retrieval.

REQ-7: System shall return an evidence-backed answer with citations and confidence.

4.3 View Idea History

4.3.1 Description and Priority

Medium priority. Enables users to review previous document analyses, queries, responses, citations, and verification information.

4.3.2 Stimulus/Response Sequences

User opens history → System fetches authorized analysis records → System displays queries, responses, source documents, citations, and confidence information.

REQ-8: System shall display past analyses belonging to the authenticated user.

REQ-9: System shall allow viewing analysis results, supporting evidence, citations, and verification status.

REQ-13: System shall allow follow-up or refined queries against authorized documents.

REQ-14: System shall support semantic search and retrieval through Qdrant.

4.4 Admin Module

4.4.1 Description and Priority

High priority. Enables administrators to manage users, documents, permissions, and system activity.

4.4.2 Stimulus/Response Sequences

Admin logs in → System verifies administrator role → Admin views authorized records → Admin performs permitted actions.

REQ-15: System shall allow admins to view document and user records.

REQ-16: System shall allow admins to manage user accounts and document access.

REQ-17: System shall restrict admin-only endpoints through RBAC.

4.5 Expert Module

4.5.1 Description and Priority

High priority. Supports expert or reviewer roles when expert-assisted analysis is enabled.

4.5.2 Stimulus/Response Sequences

Expert logs in → System verifies role → Expert accesses authorized records → Expert provides review comments or evidence assessment.

REQ-18: System shall allow experts/reviewers to log in with role-based access.

REQ-19: System shall allow authorized experts to view assigned documents or analyses.

REQ-20: Experts/reviewers shall provide structured feedback on analysis quality and evidence.

4.6 API Key Management

4.6.1 Description and Priority

Medium priority. Supports secure configuration of permitted external AI/model integrations.

REQ-21: Authorized users/admins may configure permitted external-service credentials.

REQ-22: API keys and secrets shall be validated securely and not exposed in client-side responses.

4.7 Connect with Expert

4.7.1 Description and Priority

Medium priority. Allows a user to request interaction with an authorized expert/reviewer for document-analysis guidance when enabled.

4.7.2 Stimulus/Response Sequences

User selects expert/reviewer option → System displays authorized experts → User requests interaction → System creates an authorized session.

REQ-23: System shall allow users to select authorized experts/reviewers.

REQ-24: System shall create an authorized interaction session with the selected expert/reviewer.

4.8 Testing & Validation

4.8.1 Description and Priority

High priority. Covers functional, API, UI, regression, retrieval, verification, and end-to-end validation.

REQ-25: Automated tests shall verify registration, login, document upload, querying, retrieval selection, response display, citations, and logout.

REQ-23: System shall support validation of expert/reviewer access where enabled.

REQ-26: Regression testing shall be performed after major code or model changes.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall provide responsive document processing and query handling under normal load; retrieval and reranking latency shall be monitored separately.

5.2 Safety Requirements

The application is non-critical. It shall communicate uncertainty and verification status rather than presenting unsupported claims as confirmed facts.

5.3 Security Requirements

- Passwords must use bcrypt or an equivalent secure password-hashing mechanism.
- Protected API endpoints shall require authentication.
- RBAC shall restrict sensitive endpoints and resources.
- Token expiry and secure session policies shall be enforced.
- HTTPS shall be used for frontend-backend communication.
- Rate limiting shall be applied to exposed APIs.
- Document access shall respect authorization boundaries.
- Prompt-injection and malicious-document patterns shall be detected or mitigated.

5.4 Software Quality Attributes

- Scalable for concurrent users through modular FastAPI services, PostgreSQL, Qdrant, and containerized deployment.
- Reliable through explicit error handling, retry/fallback policies, and verification-aware response generation.
- Secure through RBAC, token expiry, HTTPS, controlled file access, and secret management.
- Explainable through citations, supporting passages, verification status, and confidence information.
- Maintainable through modular service boundaries and testable retrieval/agent components.

5.5 Business Rules

- Users can only access documents, analyses, and conversations for which they have authorization.
- Administrators can manage users and system records according to assigned privileges.
- Experts/reviewers can only access records explicitly assigned or shared with them.
- External-service credentials must remain confidential and revocable.
- Generated answers should cite retrieved evidence supporting substantive claims.
- If evidence is insufficient or contradictory, the system shall show verification state and may perform corrective retrieval or request clarification.

6. Other Requirements

- Database records shall retain timestamps, user identifiers, document identifiers, and provenance information.
- Each document chunk should retain source-document and page/section metadata where available.
- Future versions may include multimodal support, additional retrieval strategies, collaboration, and improved confidence calibration.

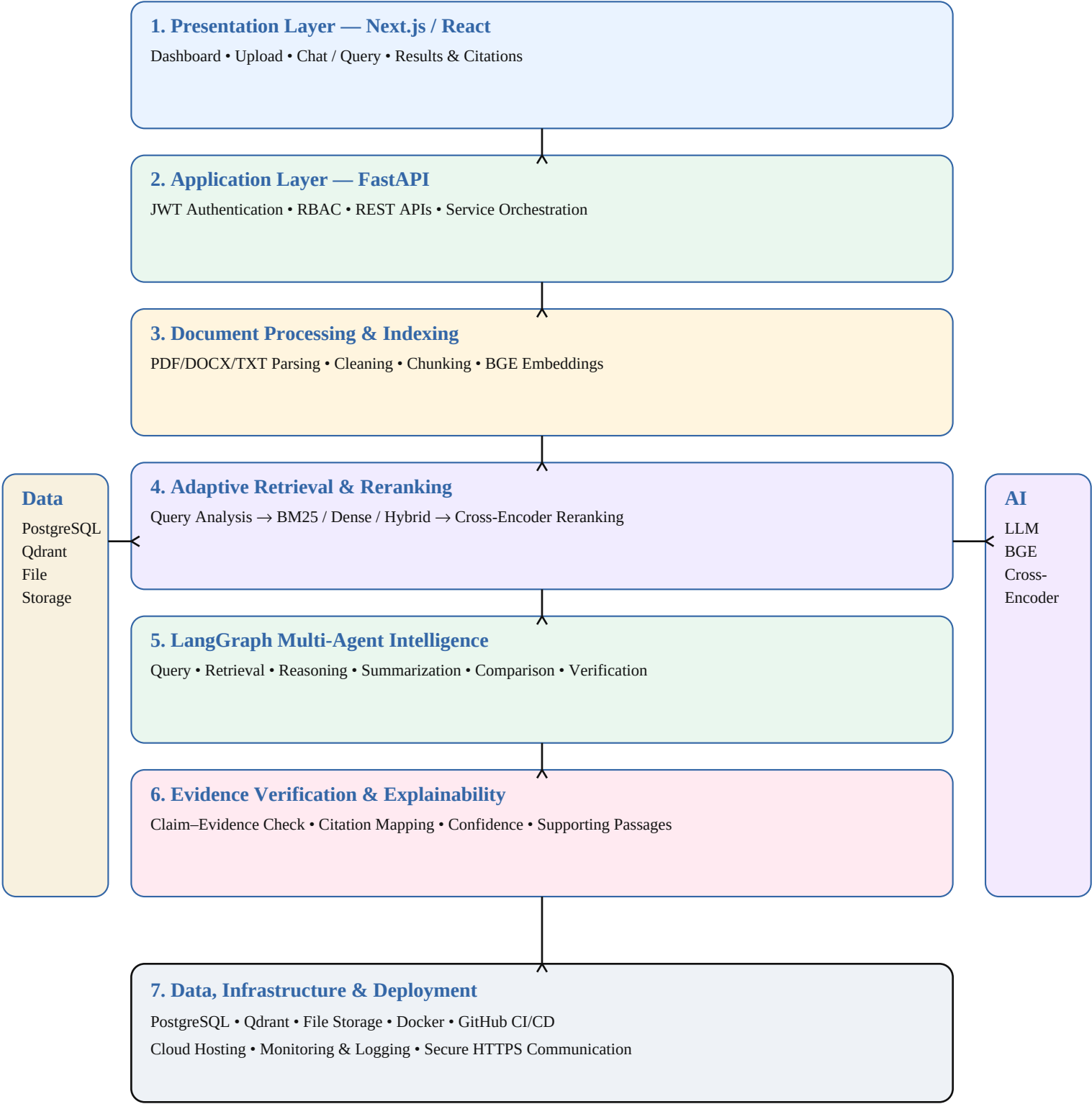
Appendix A: Glossary

RAG - Retrieval-Augmented Generation

BM25 - Best Matching 25 lexical retrieval algorithm

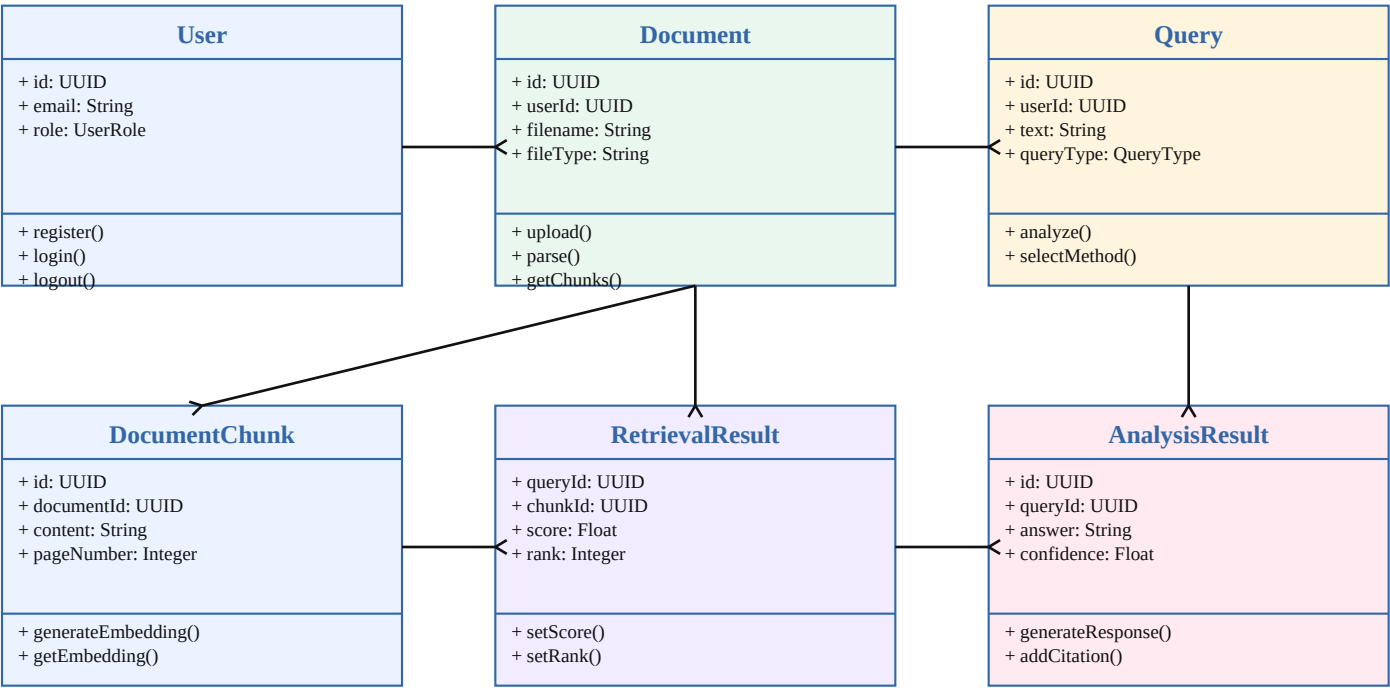
Qdrant - Vector database for embedding storage and similarity search

Appendix B: Analysis Models



Appendix B: Analysis Models

UML class model for the document-analysis workflow:



Appendix C: To Be Determined List

TBD-1: Multimodal document and image understanding.

TBD-2: Advanced confidence calibration and human-feedback evaluation.

Appendix D: Milestone Demo

A demonstration will showcase authentication, document upload, parsing and indexing, adaptive retrieval, reranking, multi-agent analysis, evidence verification, citations, confidence-aware results, document history, administrative controls, and testing/validation.